

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular/Supplementary Winter Examination – 2024

Course: B.Tech Branch : AIDS & Allied/ Computer Science and Engg (AIDS) Semester :III

Subject Code & Name: (BTAIC302) An Introduction to Artificial Intelligence

Max Marks: 60

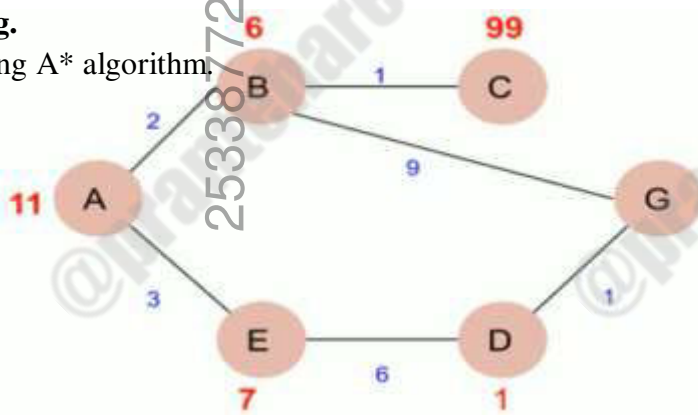
Date:07/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is NOT a characteristic of AI?	L1/CO1	1
	a) Learning from experience b) Solving problems c) Requiring constant human input d) Making decisions		
2	Which AI stage focuses on task-specific intelligence?	L2/CO1	1
	a) Artificial General Intelligence b) Artificial Narrow Intelligence c) Artificial Super Intelligence d) Artificial Specific Intelligence		
3	Who developed the Turing Test?	L1/CO1	1
	a) John McCarthy b) Alan Turing c) Marvin Minsky d) Geoffrey Hinton		
4	Which component enables agents to interact with their environment?	L2/CO1	1
	a) Sensors b) Actuators c) Both a and b d) Neither a nor b		
5	Which of the following search techniques is most likely to get stuck in a local maximum?	L3/CO2	1
	a. Best-First Search b.Hill Climbing c.Beam Search d. Tabu Search		
6	In Best-First Search, the evaluation function f(n) typically combines which two components?	L3/CO2	1
	a.Depth and branching factor b. Cost so far g(n) and heuristic estimate h(n) c.Path cost and depth d. Heuristic estimate h(n) and local maxima		
7	What does "Depth Bounded DFS" mean?	L2/CO2	1
	a) DFS with a restriction on the maximum depth it can explore b) DFS that explores all depths c) A variant of BFS with depth limitations d) DFS that prioritizes lower costs		
8	In Genetic Algorithms, what does the term "mutation" mean?	L2/CO3	1
	a) Swapping two genes randomly b) Changing the value of a gene c) Removing solutions from the population d) Creating an entirely new solution		
9	What is the key difference between A and Dijkstra's algorithm?	L2/CO4	1
	a) A* uses a heuristic to guide the search b) Dijkstra's guarantees the shortest path while A* does not c) Dijkstra's uses random exploration d) A* works only on unweighted graphs		

10	In Ant Colony Optimization, how do artificial ants simulate the process of natural ants finding the shortest path?				L2/CO4	1
	a) By generating random paths and selecting the shortest one	b) By depositing and following pheromones on paths	c) By using depth-first search to explore all possible paths	d) By employing genetic algorithms to evolve better solutions		
11	Which strategy improves the backtracking algorithm by looking ahead?				L2/CO5	1
	a) Forward Checking	b) Beam Search	c) Random Restart	d) Simulated Annealing		
12	What is "constraint satisfaction" in the broader context of AI?				L2/CO5	1
	a) Solving problems by systematically satisfying all constraints	b) Randomizing domain values	c) Optimizing heuristics without constraints	d) Generating all possible solutions without pruning		
Q. 2 Solve the following.						12
A)	Explain PEAS representation with example.				L3/CO1	6
B)	Differentiate between BFS and DFS				L2/CO2	6
Q.3 Solve the following.						12
A)	Draw suitable diagram and describe following issues of Hill Climbing Algorithm:- a) Local Maxima b) Plateaus				L2/CO2	6
B)	Explain Dijkstra Algorithm in detail				L2/CO3	6
Q. 4 Solve Any Two of the following.						12
A)	Solve the following problem using A* algorithm. Start Node : A Goal Node : G				L4/CO4	6
						
	<u>Note: The value mentioned above node is the heuristic value.</u> <u>The cost of the edge is given between the two nodes.</u>					
B)	Explain Recursive Best First Search.				L2/CO4	6
C)	Illustrate the concept of Ant Colony Optimization with neat explanation				L2/CO3	6
Q.5 Solve Any Two of the following.						12
A)	Explain the concept of Branch and Bound Algorithm				L3/CO3	6
B)	Explain Beam Search				L2/CO3	6
C)	Explain genetic algorithm.				L2/CO4	6
Q. 6 Solve Any Two of the following.						12
A)	What is Neural Network, Explain in Detail				L2/CO4	6
B)	Explain Constraint Satisfaction Problem with it's various components.				L2/CO5	6
C)	Define i) Lookahead strategies ii) Strategic retreat				L2/CO5	6
*** End ***						

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